CSC 411 Lecture 13: Probabilistic Models I

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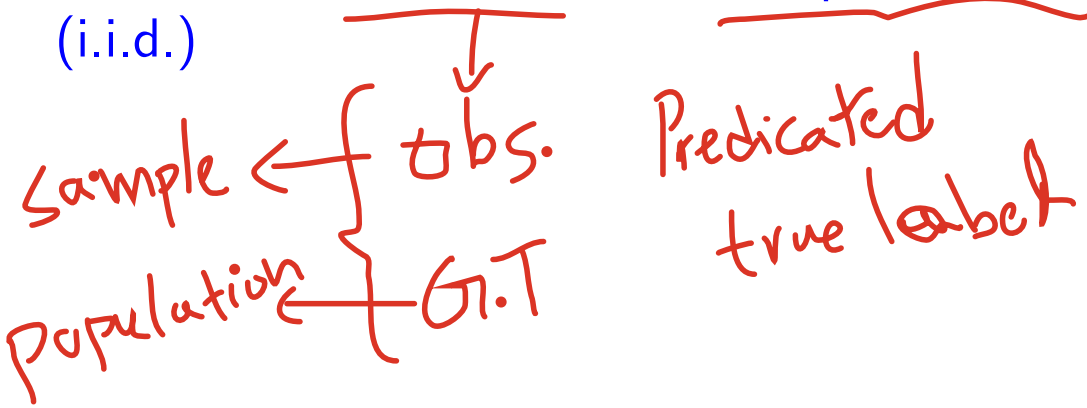
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Maximum Likelihood

- We'll shift directions now, and spend most of the next 4 weeks talking about probabilistic models.
- Today
  - maximum likelihood estimation
  - naïve Bayes

Maximum Likelihood

- Motivating example: estimating the parameter of a biased coin
  - You flip a coin 100 times. It lands heads $N_H = 55$ times and tails $N_T = 45$ times.
  - What is the probability it will come up heads if we flip again?
- Model: flips are independent Bernoulli random variables with parameter θ.
  - Assume the observations are independent and identically distributed (i.i.d.)



Maximum Likelihood

- The likelihood function is the probability of the observed data, as a function of θ.
- In our case, it's the probability of a *particular* sequence of H's and T's.
- Under the Bernoulli model with i.i.d. observations,

$$L(\theta) = p(\mathcal{D}) = \theta^{N_H}(1-\theta)^{N_T}$$

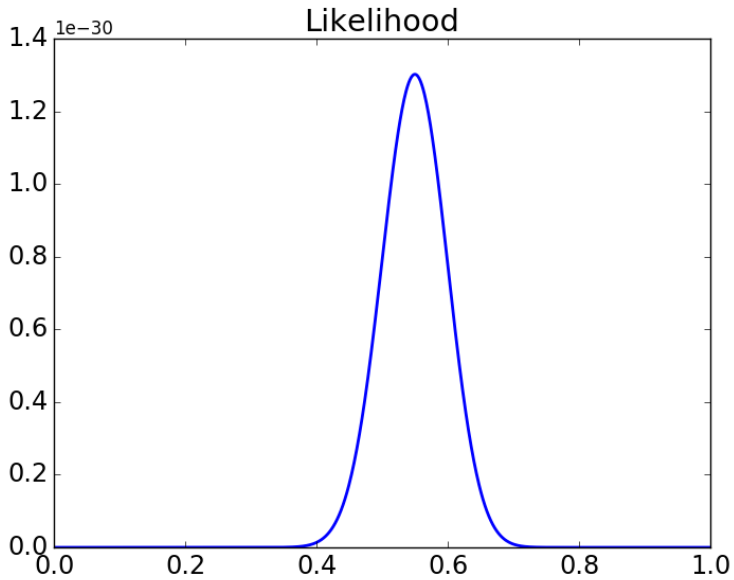
- This takes very small values (in this case, $L(0.5) = 0.5^{100} \approx 7.9 \times 10^{-31}$)
- Therefore, we usually work with log-likelihoods:

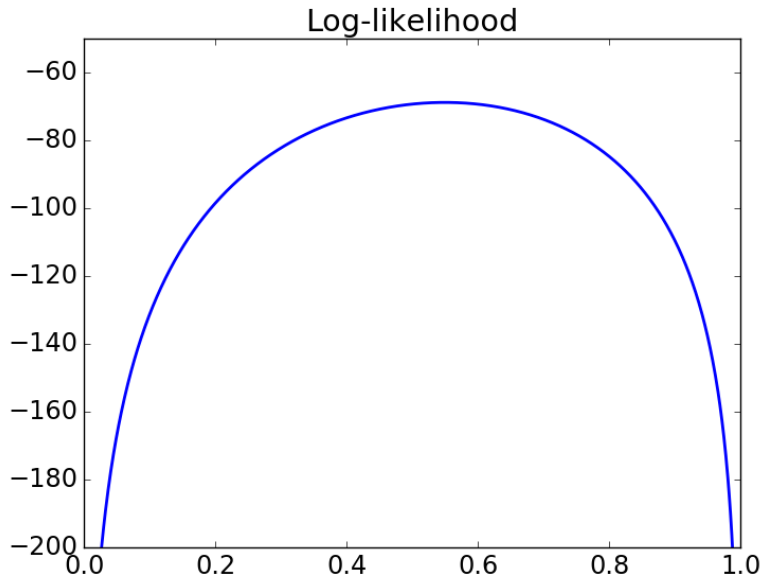
$$\ell(\theta) = \log L(\theta) = N_H \log \theta + N_T \log(1-\theta)$$

- Here, $\ell(0.5) = \log 0.5^{100} = 100 \log 0.5 = -69.31$

Maximum Likelihood

$N_H = 55,\ N_T = 45$





Maximum Likelihood

- Good values of θ should assign high probability to the observed data. This motivates the maximum likelihood criterion.
- Remember how we found the optimal solution to linear regression by setting derivatives to zero? We can do that again for the coin example.

$$\frac{\mathrm{d}\ell}{\mathrm{d}\theta} = \frac{\mathrm{d}}{\mathrm{d}\theta}\left(N_H \log\theta + N_T \log(1-\theta)\right)$$
$$= \frac{N_H}{\theta} - \frac{N_T}{1-\theta}$$

- Setting this to zero gives the maximum likelihood estimate:

$$\hat{\theta}_{\mathrm{ML}} = \frac{N_H}{N_H + N_T},$$

Maximum Likelihood

- This is equivalent to minimizing cross-entropy. Let $t_i = 1$ for heads and $t_i = 0$ for tails.

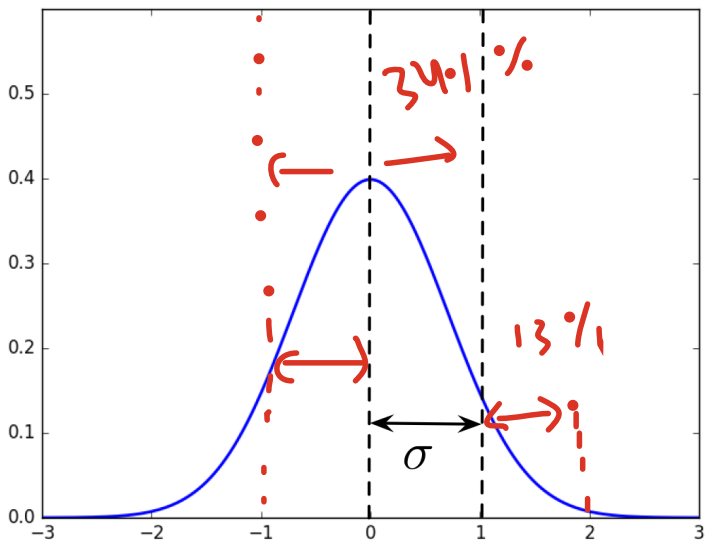
$$\begin{aligned}\mathcal{L}_{CE} &= -\sum_i t_i \log\theta - (1-t_i)\log(1-\theta) \\ &= -N_H \log\theta - N_T \log(1-\theta) \\ &= -\ell(\theta)\end{aligned}$$

Maximum Likelihood

- Recall the Gaussian, or normal, distribution:

$$\mathcal{N}(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- The Central Limit Theorem says that sums of lots of independent random variables are approximately Gaussian.
- In machine learning, we use Gaussians a lot because they make the calculations easy.



Maximum Likelihood

- Suppose we want to model the distribution of temperatures in Toronto in March, and we've recorded the following observations:

  -2.5 -9.9 -12.1 -8.9 -6.0 -4.8 2.4

- Assume they're drawn from a Gaussian distribution with known standard deviation $\sigma = 5$, and we want to find the mean μ.
- Log-likelihood function:

$$\begin{aligned}
\ell(\mu) &= \log \prod_{i=1}^{N} \left[\frac{1}{\sqrt{2\pi} \cdot \sigma} \exp\left(-\frac{(x^{(i)} - \mu)^2}{2\sigma^2} \right) \right] \\
&= \sum_{i=1}^{N} \log \left[\frac{1}{\sqrt{2\pi} \cdot \sigma} \exp\left(-\frac{(x^{(i)} - \mu)^2}{2\sigma^2} \right) \right] \\
&= \sum_{i=1}^{N} \underbrace{-\frac{1}{2} \log 2\pi - \log \sigma}_{\text{constant!}} - \frac{(x^{(i)} - \mu)^2}{2\sigma^2}
\end{aligned}$$

Maximum Likelihood

- Maximize the log-likelihood by setting the derivative to zero:

$$0 = \frac{\mathrm{d}\ell}{\mathrm{d}\mu} = -\frac{1}{2\sigma^2}\sum_{i=1}^{N}\frac{\mathrm{d}}{\mathrm{d}\mu}(x^{(i)} - \mu)^2$$

$$= \frac{1}{\sigma^2}\sum_{i=1}^{N} x^{(i)} - \mu$$

- Solving we get $\mu = \frac{1}{N}\sum_{i=1}^{N} x^{(i)}$
- This is just the mean of the observed values, or the empirical mean.

$\sigma^2 = \frac{1}{N}\sum (x^{(i)} - \mu)^2$

Maximum Likelihood

- In general, we don't know the true standard deviation σ, but we can solve for it as well.
- Set the *partial* derivatives to zero, just like in linear regression.

$$0 = \frac{\partial \ell}{\partial \mu} = -\frac{1}{\sigma^2} \sum_{i=1}^{N} x^{(i)} - \mu$$

$$\begin{aligned} 0 = \frac{\partial \ell}{\partial \sigma} &= \frac{\partial}{\partial \sigma} \left[\sum_{i=1}^{N} -\frac{1}{2} \log 2\pi - \log \sigma - \frac{1}{2\sigma^2} (x^{(i)} - \mu)^2 \right] \\ &= \sum_{i=1}^{N} -\frac{1}{2} \frac{\partial}{\partial \sigma} \log 2\pi - \frac{\partial}{\partial \sigma} \log \sigma - \frac{\partial}{\partial \sigma} \frac{1}{2\sigma} (x^{(i)} - \mu)^2 \\ &= \sum_{i=1}^{N} 0 - \frac{1}{\sigma} + \frac{1}{\sigma^3} (x^{(i)} - \mu)^2 \\ &= -\frac{N}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^{N} (x^{(i)} - \mu)^2 \end{aligned}$$

$$\hat{\mu}_{\mathrm{ML}} = \frac{1}{N} \sum_{i=1}^{N} x^{(i)}$$

$$\hat{\sigma}_{\mathrm{ML}} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (x^{(i)} - \mu)^2}$$

Maximum Likelihood

- Sometimes there is no closed-form solution. E.g., consider the gamma distribution, whose PDF is

$$p(x) = \frac{b^a}{\Gamma(a)} x^{a-1} e^{-bx},$$

  where Γ is the gamma function, a generalization of the factorial function to continuous values.
- There is no closed-form solution, but we can still optimize the log-likelihood using gradient ascent.

Maximum Likelihood

- So far, maximum likelihood has told us to use empirical counts or statistics:
  - **Bernoulli:** $\theta = \frac{N_H}{N_H + N_T}$
  - **Gaussian:** $\mu = \frac{1}{N}\sum x^{(i)}$, $\sigma^2 = \frac{1}{N}\sum (x^{(i)} - \mu)^2$
- This doesn't always happen; the class of probability distributions that have this property is exponential families.

Maximum Likelihood

We've been doing maximum likelihood estimation all along!

- Squared error loss (e.g. linear regression)

$$p(t|y) = \mathcal{N}(t; y, \sigma^2)$$
$$-\log p(t|y) = \frac{1}{2\sigma^2}(y-t)^2 + \text{const}$$

- Cross-entropy loss (e.g. logistic regression)

$$p(t=1|y) = y$$
$$-\log p(t|y) = -t \log y - (1-t)\log(1-y)$$